Bioresource Technology 247 (2018) 881–889

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

Influence of pretreatment conditions on lignocellulosic fractions and methane production from grape pomace

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ARTICLE INFO

Keywords:
Anaerobic digestion
Biochemical Methane Potential
Pretreatments
Grape pomace
Lignocellulosic fractions

ABSTRACT

The lignocellulosic structure of grape pomace requires the use of pretreatments facilitating microbial decomposition of the matter and enhancing methane production. In this study, the effects of various pretreatments (freezing, alkaline treatment using NaOH and NH_3, acid treatment using HCl, ultrasounds and pulsed electric fields) were examined in batch mode. The highest methane production (0.178 $Nm^3\ kg^{-1}$ of COD) was attained after alkaline treatment with 10% NaOH w/w dry basis, at 20 °C and for 24 h. This result is due to the degradation of more than 50% of lignin and about 22% of cellulose present in grape pomace. The coupling of this pretreatment with freezing at −20 °C exhibited the highest methane production of 0.2194 ± 0.0007 $Nm^3\ kg^{-1}$ of COD. When applied to a larger scale continuous digester, this coupled pretreatment increased methane production by about 27%, compared to the untreated samples, promoting the green valorization of the biomass.

1. Introduction

Nowadays, the production of bioenergy from agricultural biomass has become a global concern. Lignocellulosic residues are valuable raw materials for the production of methane through Anaerobic Digestion (AD), thus facilitating the transition to renewable green energy sources (Kumar and Sharma, 2017; Monlau et al., 2013a). In fact, biomethane is a versatile energy source because it can be used to produce heat, electricity combined with heat (cogeneration) or biofuel. Compared to most liquid biofuels, biomethane showed better performance in terms of both agricultural land area efficiency and life cycle emissions (Börjesson and Mattiasson, 2008).

The grape is one of the most cultivated plant crops in the world (FAOSTAT, 2014). According to an estimation reported by the International Organization of Vine and Wine, its total production amounted 75.8 million tons with 267 million hectoliters of wine in 2016 (OIV, 2017). The European Union has participated in nearly 60% of world production with Italy, France and Spain as the three main producer countries. The winemaking process generates large quantities of solid waste, especially grape pomace (GP), its major by-product (62% of the generated waste) (Schieber et al., 2001). We have shown previously that GP is a promising potential source of methane, targeting consequently two issues: green energy production as well as waste management and valorization (El Achkar et al., 2016).

The main challenge in the use of lignocellulosic crops, such as GP, for methane production, is their structure and composition (Monlau et al., 2012). The solid matter of GP is mainly composed of cellulose, hemicelluloses and lignin. The quite low anaerobic biodegradability of the GP is likely due to the high amount of lignin and cellulose in its dry matter. In fact, lignin is the most resistant to microbial degradation and oxidation (Xu et al., 2014), whereas cellulose is usually degradable under anaerobic conditions in its pure form (Carlsson et al., 2012). However, when combined with lignin, cellulose shows limited accessibility to hydrolytic enzymes and anaerobic microorganisms. On another note, carbon in hemicellulose is easily available and thus readily degradable by microorganisms (Mottet et al., 2010). Therefore, GP pretreatment before AD seems necessary to alter the structural properties of lignin and cellulose, reducing the complexity of lignocellulosic fractions. Accordingly, the development of innovative and low-cost pretreatments methods is required to facilitate the microbial decomposition of the matter, to increase the GP degradation and hence the methane yield.

Different pretreatment methods have been widely studied by Hendriks and Zeeman (2009), Carlsson et al. (2012) and Carrere et al. (2016) to facilitate the enzymatic hydrolysis and the consequent methane production from organic waste. To the best of our knowledge,

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http://dx.doi.org/10.1016/j.biortech.2017.09.182
Received 27 July 2017; Received in revised form 25 September 2017; Accepted 26 September 2017
Available online 28 September 2017
0960-8524/

there are no studies evaluating the effects of different pretreatments on the methane production and the chemical composition of GP. In this study, we have assessed the impact of selective treatment methods on GP AD. These pretreatments, commonly used on a wide range of lignocellulosic substrates, include freezing pretreatment, chemical pretreatments (acid and alkaline) and physical pretreatments (ultrasounds and pulsed electric fields) (Carlsson et al., 2012; Carrere et al., 2016; Yadvika et al., 2004). Biological pretreatments (using fungi and enzymes) are not considered here, mainly due to the high loss of carbohydrates, the long treatment time and their high cost (Sambusiti et al., 2012). Moreover, and despite their efficiency in solubilizing crops, thermal pretreatments were not inspected since they are very energy intensive, particularly when compared to freezing process (Stabnikova et al., 2008). Studying freezing process is of a great interest because it may occur during outdoor storage of GP in winter conditions.

In sum, to the best of our knowledge, this is the first study investigating the influence of various pretreatments on GP methane potential and structural features. The main objectives consist in: (i) determining the effects of various pretreatments (freezing, alkaline treatment using NaOH and NH_3, acid treatment using HCl, ultrasounds (US) and pulsed electric fields (PEF)) on GP methane production in batch digesters; (ii) investigating the impact of these pretreatments on the biodegradability of the main lignocellulosic components; (iii) validating the most efficient pretreatment, in terms of GP methane production, on a larger scale continuous digesters.

2. Materials and methods

2.1. Materials

The GP variety "Chenin blanc", was obtained from a wine company (Domaine Des Acacias, Layon, France). The grapes were carefully harvested at maturity during the 2015 vintage and fresh samples of GP were collected immediately after the pressing operation and transported to the laboratory. On arrival, stalks fragments were removed manually from the set of GP. The raw material was stored at +6 °C until utilization. An active inoculum was collected from the digester of the waste-water treatment plant of Saint-Brieuc city in France, operating under mesophilic conditions. The particulate matter (> 500 mm) was removed from the inoculum by passing through sieve to ensure better homogeneity and to improve the reproducibility of the experimental tests.

2.2. Analyses

Total Solids (TS) content were determined by dry weight in oven at 105 °C until constant weight. Afterwards, Volatile Solids (VS) were determined in a muffle furnace after 4 h at 550 °C. Total Chemical Oxygen Demand (COD) was measured using Merck COD Spectroquant® test, range 500–10,000 mg L^{-1}, and a spectrophotometer NOVA 60 (Merck, Germany). For this last analysis, the GP was ground using a blender (Waring blender 8011EG, Waring Commercial, USA). After 2 h at 150 °C, it is considered that all organic matter is oxidized with the hot sulfuric solution of potassium dichromate.

2.3. Van Soest fractionation

The structural polysaccharides and lignin fractions were determined using the method of Van Soest et al. (1991). A detailed description of the protocol used was provided in a previous paper (El Achkar et al., 2016). All reagents were supplied by Merck (Darmstadt, Germany) and all analytical determinations were performed in triplicate.

2.4. Pretreatments of GP

2.4.1. Freezing

Some of the fresh GP were treated using a freezing/thawing technique. First set of GP was frozen at −20 °C and the second at −80 °C, for 48 h. Afterwards, samples were thawed for 24 h at 6 °C.

2.4.2. Chemical treatments (NaOH, NH_3 and HCl)

The effects of chemical treatments on the methane production of GP were evaluated using NaOH, NH_3 and HCl, separately. These pretreatments were carried out in 250 mL flasks, closed with rubber septa. In each flask, GP samples are introduced with a total solid load of 50 g L^{-1} (Monlau et al., 2013b). Experiments conditions were determined according to some literature suggestions on agricultural substrates (Monlau et al., 2012; Sambusiti et al., 2012). Therefore, fresh GP samples were soaked in chemical agent concentration of 6% and 10% w/w dry basis. To ensure sufficient mixing, the flasks were continuously shacked at 150 rpm using an orbital shaker (IKA KS 260 basic, Labtek, Staufen, Germany). Chemical treatments were carried out at ambient temperature (20 ± 1 °C), for 24 h. At the end of the treatment, the pretreated samples were neutralized with a concentrated HCl or NaOH solutions to pH 7.5 prior to BMP test.

2.4.3. PEF treatment

The treatment chamber consists of a plexiglass cylinder (6 cm of height) with two steel electrodes (inner diameter of 6.5 cm). The electrodes were connected to the PEF system which consists of a high voltage power supply (2.5 kV, 0.24A) (SR2.5-P-600, Technix, France), a pulse generator (TGP 110–10 MHz Pulse Generator – TTI Thurlby Thandar Instruments, France) used to control the PEF treatment protocol by regulating the pulse duration t_i (10^{-5}–10^{-4} s) and the frequency (24–240 Hz). To check the treatment protocol, an oscilloscope (OX 8022 – 20 MHz Differential, Metcix, France) has been used. The high voltage power supply and the pulse generator were connected to a modulator (AHTPM2.5, Effitech, Pau, France) which combines the generated high voltage with the pulse protocol to obtain the desired PEF. The measurement of the electrical conductivity, before and after the treatment, was conducted using an LCR meter (U1733C, Agilent, Les Ulis, France).

The PEF treatment effect, resulting in tissue damages, was estimated from the disintegration index (Z), based on the variation of the electrical conductivity before and after treatment (Lebovka et al., 2002).

$$Z = \frac{(\sigma_m - \sigma_i)}{(\sigma_e - \sigma_i)} \tag{1}$$

where σ_m is the measured electrical conductivity value after the pause duration (Δt) between trains, σ_i and σ_e refer to the conductivities of the untreated (initial) and totally electroporated sample, respectively. Eq. (1) gives Z = 0 for an intact sample and Z = 1 for a completely damaged tissue; here, three levels of PEF treatment were determined: Z_0, $Z_{0.5}$ and Z_1.

The specific energy input W for each level of PEF treatment was calculated by summation of power consumption for each train and divided by the mass m of the sample in the treatment chamber:

$$W_{PEF} = \sum_{1}^{N} U\ I\ n\ t_i/m \tag{2}$$

where U is the PEF applied voltage, I is the treatment's current intensity. As I varied during the treatment from a train to another, the specific energy for each train was obtained with the I value calculating from the measured conductivity of the sample. A moderate intensity of PEF of 3600 V cm^{-1} was used. Rectangular monopolar trains (n = 240 pulses) with $50 \cdot 10^{-6}$ s pulse duration (t_i) were applied and the number of trains (N) was 9 for $Z_{0.5}$ and 14 for Z_1. PEF treatment specific times (t_{PEF}) were 0.36 s and 1 s, for $Z_{0.5}$ and Z_1, respectively. To neglect the thermal effects, the elevation of temperature was kept below 5 °C by adjusting a pause time Δt (5 min) between the trains. Energy inputs were 41 kJ kg^{-1} and 153 kJ kg^{-1}, for $Z_{0.5}$ and Z_1, respectively.

2.4.4. US treatment

The disintegration by US was performed with an ultrasonic processor (BPAC, Vern-Sur-Seiche, France) with an operating frequency of 50 kHz and an effective power output of 60 W. The sonication bath, with a capacity of 2 L, has the internal dimensions of 15 cm length, 13.5 cm breadth and 10 cm height. The total internal body is made up of stainless steel. The temperature of GP samples was controlled so as not to exceed 25 °C. The GP is put in 0.25 L bottles, with a total solid load of 50 g L^{-1}, as mentioned previously.

To assess the degree of disintegration, the COD of GP supernatant was determined. The degree of disintegration (DD_{COD}) is calculated as the ratio between soluble COD increase due to sonication and the total COD (Braguglia et al., 2008):

$$Z = DD_{COD} = \left[\frac{COD_{sol,\ sonicated} - COD_{sol,\ untreated}}{COD_{sol,\ max} - COD_{sol,\ untreated}}\right] \quad (3)$$

where $COD_{sol,\ sonicated}$, $COD_{sol,\ untreated}$, are soluble COD (mg L^{-1}), of the sonicated and of the untreated sample, respectively, and $COD_{sol,\ max}$ is COD of the completely disintegrated sample.

From this equation, two different levels of disintegration of GP were determined. $Z_{0.5}$ corresponds to half-disintegration of the GP by applying US treatment duration of 40 min. Z_1 corresponds to maximum disintegration with a treatment duration of 70 min.

The specific US energy input W_{US} was calculated as follows:

$$W_{US} = \frac{P_{US} \times t_{US}}{m} \quad (4)$$

where P_{US} is the power of the US generator (0.06 kW), m is the mass of GP + water mixture (kg) and t_{US} is the US treatment duration (s). In this study, energy inputs were 400 kJ kg^{-1} and 700 kJ kg^{-1}, for $Z_{0.5}$ and Z_1, respectively.

2.5. Automatic methane potential test system

The maximal methane production during AD experiments was determined using an Automatic Methane Potential Test System (AMPTS, Bioprocess Control, Lund, Sweden). This system measures the methane production of different samples over time. Fresh inoculum and substrate were added to 500 mL incubation bottles at the same COD inoculum to substrate ratio of 3:1 (El Achkar et al., 2016). Bottles were incubated at 37 °C until the methane production has stopped. All tests and blanks were carried out in triplicate, and the net values of methane production and yield were obtained by subtracting the endogenous production of the blank bottles. The biological methane potential (BMP) was calculated from the volume of methane produced divided by the quantity of the sample based on COD introduced. The biodegradability (%) was determined by dividing BMP by the theoretical maximum methane yield (0.35 Nm^3 kg^{-1} of COD) under standard conditions of temperature and pressure (273 K and 101.5 kPa). At the end of the experiments, the measurements of pH, alkalinity and total volatile acidity (TVA) were carried out as described previously (El Achkar et al., 2016).

2.6. Kinetic study

The methane potential test can be used to estimate the overall hydrolysis constant k. In descriptive terms, this reaction step is typically simplified and reduced to first-order kinetics (Eastman and Ferguson, 1981).

$$BMP = BMP_0(1 - e^{-kt}) \quad (5)$$

where BMP is the cumulative methane yield, BMP_0 is the maximum methane yield of the substrate, k (days^{-1}) is the apparent kinetic constant and t (days) is the time. The adjustment by nonlinear regression of the pairs of experimental data (BMP, t) using GraphPad Prism® (GraphPad, San Diego, CA; USA) allows the calculation of the apparent kinetic constant k and R^2. Root Mean Square Prediction Error (RMSPE) was calculated as mentioned by (El-Mashad, 2013) in order to validate the studied model. It represents the deviation between experimental and estimated methane production. High values of the coefficient of determination ($R^2 > 0.98$ in all cases) and small RMSPE values (1–6% of the total methane production for each GP variety) indicate that the first-order kinetics model describes and fits the data well.

2.7. Methane production in continuous mode

The system for measuring methane production in continuous mode contains three main units. The incubation unit is a thermostatic water bath (37 °C) which contains four digesters of 2 L (Bioprocess Control, Lund, Sweden). Each digester is equipped with two sealable tubes; The first for drawing off the digestate and the second for supplying the GP. Digesters are stirred manually for one minute, twice a day, to ensure adequate mixing. The produced biogas passes through a tube connecting the incubation unit to the CO_2 fixing unit (6 M NaOH solution). Thus, the CH_4 released from this unit is measured with a gas meter (Type MGC-1 PMMA, Ritter, Germany) with a measurement range between 1 mL h^{-1} to 1 L h^{-1}.

Feeding and draining operations were carried out daily using a hydraulic retention time (HRT) of 20 days and an organic loading rate of 3.7 kg COD m^{-3} day^{-1}. Therefore, using a plastic syringe, 100 mL of the digestate were taken and a similar volume of GP was introduced into the digesters. Digesters operated for 70 days (3.5 the HRT duration) to record stability in methane production and physicochemical parameters.

GP was grinded by means of a blender (Waring blender 8011EG, Waring Commercial, USA) in a total volume of 0.8 L for 10 min and with a speed of 22,000 rpm. The particle size distribution was determined by laser light scattering (Mastersizer 2000, Malvern Instruments, UK). The particle size distribution is calculated as volume based in terms of equivalent spheres (range 0.02–2000 μm).

2.8. Statistical analysis of experimental data

Significant statistical differences between means at 0.01% and 0.05% levels were determined according to Tukey's test. Statistical tests were conducted using SPSS (SPSS software, V20, SPSS Inc., Chicago, USA).

3. Results and discussion

3.1. Impact of pretreatments on methane production

GP variety "Chenin Blanc" has a TS content of 454 ± 14 g kg^{-1}, a VS content of 350 ± 10 g kg^{-1} and COD of 610 ± 20 g O_2 kg^{-1}. The inoculum used had a pH value of 7.5, a TS content of 56 ± 3 g kg^{-1}, a VS content of 35 ± 4 g kg^{-1} and COD of 52.0 ± 0.1 g O_2 kg^{-1}. The microorganisms present in the inoculum and their characteristics are shown to affect significantly the methane yield in the AD process. It should be emphasized that the same inoculum was used in all experiments, so the possibility of being a source of comparison error must be rejected. For all samples, the cumulative production of methane increases exponentially during the initial phase until day 8–10 after the starting of the experiments, due to components that are easily hydrolysable by microorganisms. A plateau is then obtained indicating a depletion of biodegradable matter. The cumulative methane yield of the untreated GP at the end of test were 0.1305 ± 0.0035 Nm^3 kg^{-1} of COD.

For better monitoring of the process, the physicochemical

Table 1
Physico-chemical parameters of the medium, at the end of the batch experiments.

Treatment conditions	Final pH	Final TVA (g eq. $CH_3COOH\ L^{-1}$)	Final Alkalinity (g eq. $CaCO_3\ L^{-1}$)
Freezing at −80 °C	7.2	0.15	3.2
Freezing at −20 °C	7.2	0.15	3.2
6% NaOH	7.5	0.14	3.4
10% NaOH	7.6	0.13	3.6
6% NH_3	7.2	0.16	3.0
10% NH_3	7.2	0.16	3.1
6% HCl	7.2	0.18	3.0
10% HCl	7.2	0.18	2.9
US $Z_{0.5}$	7.2	0.16	3.2
US Z_1	7.2	0.16	3.3
PEF $Z_{0.5}$	7.3	0.17	3.3
PEF Z_1	7.2	0.17	3.2
Freezing at −20 °C + 6% NaOH	7.5	0.13	3.5
Freezing at −20 °C + 10% NaOH	7.5	0.13	3.5

parameters (pH, Total Volatile Acidity (TVA) and alkalinity) were measured at the end of the batch experiments. The pH of the medium decreased from 7.5 to a minimum of 7.2 (Table 1). Total Volatile Acidity (TVA) increased from 0.11 to a maximum of 0.18 g eq. $CH_3COOH\ L^{-1}$ and Alkalinity decreased from 3.7 to a maximum of 2.9 g eq. $CaCO_3\ L^{-1}$ and, by the end of experiments. It should be noted that, for a successful AD process, the alkalinity must be maintained above 2 g eq. $CaCO_3\ L^{-1}$ to avoid drastic drops in pH and TVA, must be less than 2 g eq. $CH_3COOH\ L^{-1}$ to prevent the inhibition of methanogenic bacteria (Yadvika et al., 2004). Here, the results attest that methane production stops due to the depletion of the biodegradable COD and not to the medium acidification or process inhibition.

3.1.1. Freezing

First, we conducted AD on GP, frozen at −80 °C and −20 °C, recording at the end of the test cumulative methane yields of 0.1508 ± 0.0007 and 0.1696 ± 0.0003 $Nm^3\ kg^{-1}$ of COD, respectively (Fig. 1). Significant increases in the BMP_0 of 16% and 30% were observed in comparison to the raw samples. By against, no significant increases in first-order kinetics were noted (Table 2). These results demonstrate that freezing at −20 °C is more effective than freezing at −80 °C in improving the methane production of GP. This increase can be explained by the impact of the freezing rate on the cells and their organoleptic consequences (Rabin et al., 1996; Thomashow, 1998). In fact, during slow freezing (at −20 °C), deterioration caused by ice can be explained by the rigidity and the large size of crystals which generate mechanical pressure damaging the cellular structure. In addition, slow freezing leads to the dehydration of the cells via osmosis, which results in a weakening of the cellular tissue and a loss of its turgidity. However, the fast freezing kinetics (at −80 °C) create a multitude of small crystals in the cell, without denaturation of cellular structures. Moreover, rapid freezing process enables also a significant reduction in dehydration compared to slow freezing. As a result, GP frozen at −20 °C was more damaged than that frozen at −80 °C, which led to an increase in its biodegradability and methane production.

3.1.2. Alkaline treatment with NaOH

Alkaline treatment of GP with NaOH has recorded the highest methane production compared to all other pretreatments. Cumulative methane yields at the end of test were 0.1571 ± 0.0005 and 0.1780 ± 0.0004 $Nm^3\ kg^{-1}$ of COD for GP treated with 6% NaOH and 10% NaOH, respectively (Fig. 2). Significant increases in the BMP_0 of 20% and 36% were therefore highlighted in comparison with untreated samples. In addition, higher hydrolysis kinetics were recorded with 6% and 10% NaOH, corresponding to 40% and 71% increase, respectively (Table 2). In fact, the effect of the alkaline treatment using NaOH on methane production was evaluated in numerous studies. The treatment of rice straw with 1% NaOH, for 3 h at 20 °C, improved methane production by 34% (Shetty et al., 2017). Comparable augmentations were registered for various vegetal biomasses treated with 5–8% NaOH, for different time durations; we mention mainly corn stover and asparagus stems (Chen et al., 2014; Zhu et al., 2010). A particular higher increase in BMP_0 by almost 94% was observed for poplar processing residues treated with 8% NaOH, for 4 days at 20 °C (Yao et al., 2013).

3.1.3. Alkaline treatment with NH_3

Alkaline treatment of GP with NH_3 did not affect the methane production yields in comparison with the raw GP. The cumulative methane yields at the end of test were 0.1295 ± 0.0005 and 0.1304 ± 0.0007 $Nm^3\ kg^{-1}$ of COD for GP treated with 6% NH_3 and 10% NH_3, respectively (Table 2). Furthermore, no significant increases in first-order kinetics were observed. It should be noted that the AD process was not inhibited which therefore proves that the presence of ammonia was not a barrier to the degradation of GP by anaerobic microorganisms. Similar results were obtained after alkaline treatment of wheat straw with NH_3; they showed that methane production was not significantly affected (Nordmann, 2013).

3.1.4. Acid pretreatment with HCl

Regarding the acid pretreatment using HCl, the cumulative methane yields at the end of test were 0.1352 ± 0.0003 and 0.1375 ± 0.0003 $Nm^3\ kg^{-1}$ of COD for GP treated with 6% HCl and 10% HCl, respectively. BMP_0 showed a significant increase of 4% and 5%, respectively, in comparison with the raw GP, as where the hydrolysis constant increased by 29% (Table 2). Such positive impact was also observed in many studies evaluating the effect of HCl on AD of many lignocellulosic substrates, such as sugarcane bagasse and coconut

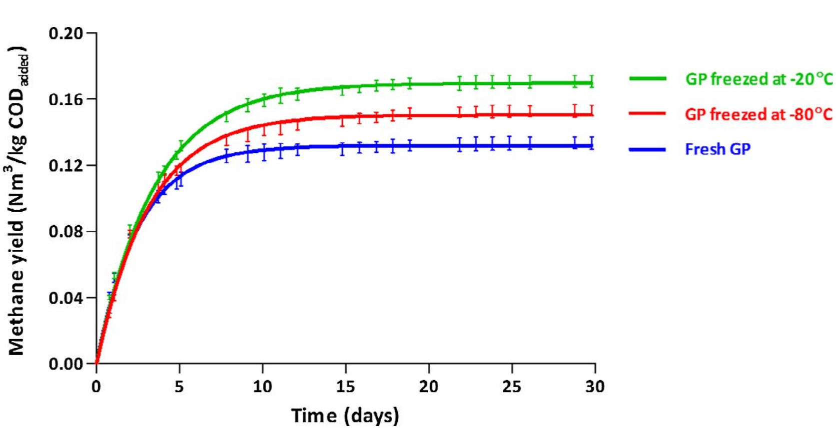

Fig. 1. Cumulative methane production of the GP after freezing pretreatment. Lines correspond to a first-order model and bars represent the average standard deviation of the experimental data.

Table 2
Methane production (BMP), hydrolysis kinetics constant (k) and anaerobic biodegradability (BD) for selected conditions of raw and pretreated GP. Values corresponds to mean ± standard deviation of measurement performed in triplicate.

Conditions	BMP_0 (Nm^3 CH_4 kg COD^{-1})	R^2	RMSPE (Nm^3)	Δ BMP_0 (%)	k (d^{-1})	Δ k (%)	BD (%)
Untreated GP	0.1305 ± 0.0035	0.993	0.0035	–	0.2798 ± 0.0011	–	37
Freezing at −80 °C	0.1508 ± 0.0007	0.998	0.0004	+16**	0.2858 ± 0.0029	+2	43
Freezing at −20 °C	0.1696 ± 0.0003	0.998	0.0005	+30**	0.3051 ± 0.0070	+9	48
6% NaOH	0.1571 ± 0.0005	0.999	0.0018	+20**	0.3928 ± 0.0060	+40**	45
10% NaOH	0.1780 ± 0.0004	0.999	0.0021	+36**	0.4793 ± 0.0091	+71**	51
6% NH_3	0.1295 ± 0.0005	0.993	0.0064	0	0.3099 ± 0.0121	+11	37
10% NH_3	0.1304 ± 0.0007	0.990	0.0068	0	0.3107 ± 0.0106	+11	37
6% HCl	0.1352 ± 0.0003	0.998	0.0044	+4*	0.3597 ± 0.0058	+29**	39
10% HCl	0.1375 ± 0.0003	0.998	0.0048	+5*	0.3620 ± 0.0085	+29**	39
US $Z_{0.5}$	0.1312 ± 0.0006	0.993	0.0054	+1	0.3497 ± 0.0131	+25**	37
US Z_1	0.1486 ± 0.0003	0.998	0.0052	+10**	0.3773 ± 0.0181	+35**	43
PEF $Z_{0.5}$	0.1315 ± 0.0015	0.989	0.0082	+1	0.3062 ± 0.0215	+9	38
PEF Z_1	0.1356 ± 0.0020	0.986	0.0086	+4*	0.3192 ± 0.0221	+14*	39
Freezing at −20 °C + 6% NaOH	0.2060 ± 0.0009	0.999	0.0021	+58**	0.3055 ± 0.0068	+9	59
Freezing at −20 °C + 10% NaOH	0.2194 ± 0.0007	0.997	0.0019	+68**	0.3110 ± 0.0058	+11	63

* The increase is significant at value 0.05.
** The increase is significant at value 0.01.

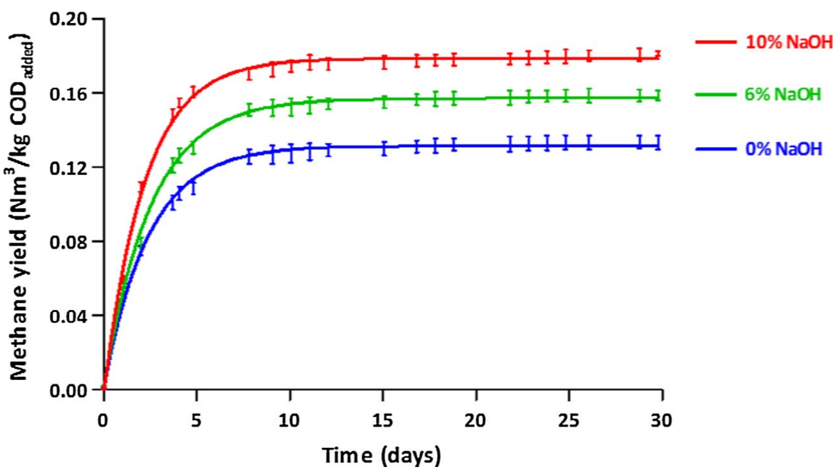

Fig. 2. Cumulative methane production of the GP after alkaline pretreatment using NaOH. Lines correspond to a first-order model and bars represent the average standard deviation of the experimental data.

fibers (Kivaisi and Eliapenda, 1994). On the other hand, treatment of maize plants with 2% HCl, for 24 h at 20 °C did not show any effects on methane production which remained constant (Pakarinen et al., 2011).

3.1.5. PEF pretreatment

The effects of PEF pretreatments on the AD of GP were also investigated. The Z_1 level of PEF (W = 153 kJ kg^{-1}) increases the cumulative methane yield and the hydrolysis constant by 4% and 14%, respectively (Table 2). This technology has been mainly used as a pretreatment upstream of AD carried out on sludge (Carlsson et al., 2012). For example, PEF applied to different waste activated sludge with higher energy inputs led to very high increases in biogas production (Lee and Rittmann, 2011).

3.1.6. US pretreatment

The Z_1 level sonication of GP (W = 700 kJ kg^{-1}) engendered an increase of 10 and 35% in the corresponding BMP_0 and hydrolysis constant (Fig. 3 and Table 2). Usually, US with low frequencies are largely coupled to AD of sewage sludge (Pérez-Elvira et al., 2009). Recently, several research teams have studied US effects on solid waste AD. For example, the sonication of maize straw and dairy manure in codigestion, using an energy of 285 kJ kg^{-1} of TS, has increased biogas production by about 70% (Zou et al., 2016).

3.1.7. Alkaline pretreatment using NaOH combined with freezing at −20 °C

Overall, our previous findings highlight that freezing treatment at −20 °C and alkaline treatment using 10% NaOH led, separately, to maximal increase in BMP_0. To maximize methane production from GP, we assessed the effects of the combination of these two pretreatments. It displayed the highest increase in BMP_0 (58% and 68%) among all pretreated samples (Table 2). A maximum GP biodegradability of 63% was reached for GP frozen at −20 °C and treated with 10% NaOH; listing this combination of pretreatments at the top of our selected ones.

3.2. Impact of selected pretreatment conditions on GP composition

After choosing conditions exhibiting the highest improvement in methane production for each pretreatment (freezing at −20 °C, 10% NaOH, 10% NH_3, 10% HCl, PEF Z_1, US Z_1 and 10%NaOH + freezing at −20 °C), we have analyzed the composition of raw GP and treated samples, using Van-Soest fractionation method.

Table 3 indicates that lignin, cellulose and hemicellulose are the three main components of lignocellulosic biomass representing about 49.5% of the VS in raw GP. In addition, we demonstrate that an increase in soluble matter, resulting from the removal of hemicellulose, cellulose and lignin, is positively correlated with an improvement of GP methane yield (R^2 = 0.81); our results being consistent with previous study (Jackowiak et al., 2011). Moreover, another significant positive correlation was found between the increase in soluble matter and

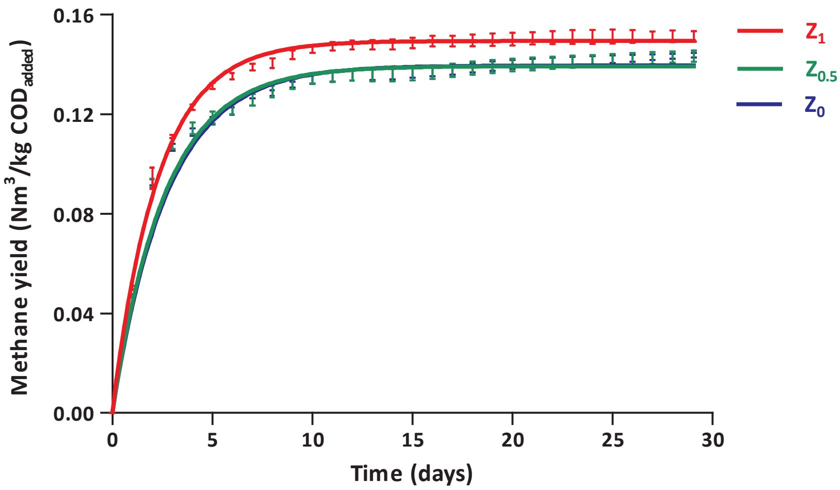

Fig. 3. Cumulative methane production of the GP after US treatment. Z_0 corresponds to the untreated GP, $Z_{0.5}$ to the half-disintegrated GP and Z_1 to the completely disintegrated GP. Lines correspond to a first-order model and bars represent the average standard deviation of the experimental data.

hydrolysis kinetic constants ($R^2 = 0.94$); the higher the soluble matter, the higher the kinetic constants. These results support another study evaluating the effects of seven types of thermo-chemical pretreatments on the AD of sunflower stalks, highlighting important correlation between the sum of solubilized proteins, hemicellulose, cellulose and lignin and the hydrolysis kinetic constants ($R^2 = 0.91$) (Monlau et al., 2012).

Freezing of the GP at −20 °C did not alter its lignocellulosic composition (Fig.4). However, it enhanced as previously mentioned, its methane production by almost 30% (Table 2). In fact, it has been shown that high cell disintegration can be reached during freezing at low temperature without changing the chemical composition of the cell matrix (Thomashow, 1998). Moreover, grape freezing is usually used to promote the release of phenolic compounds from the skins. It causes the burst of the cells, breaking their membranes and thus releasing preferentially water-soluble compounds (Heredia et al., 2010). Therefore, the increase in methane production is directly linked to the release of the soluble compounds while lignocellulosic fractions remain intact (Jackowiak et al., 2011). For these reasons, freezing of food waste is considered crucial to facilitate the hydrolytic process in anaerobic digesters ensuring faster supply of nutrients to microorganisms (Stabnikova et al., 2008).

In our study, alkaline treatment with 10% NaOH was the only treatment that altered the three components of the lignocellulosic biomass, justifying the highest enhancement in methane production. Significant reduction yields of 27.9%, 22.3% and 52.3% were obtained, for hemicellulose, cellulose and lignin, respectively (Fig. 4). In fact, Sodium Hydroxide pretreatment of lignocellulosic biomass has been shown to modify lignin structure, increasing the enzymatic accessibility to cellulose and hemicellulose fractions (Cheng et al., 2010; Sills and Gossett, 2011). Here, hemicellulose content decreased more than that of cellulose whereas lignin removal rate was much higher than both; this is in agreement with another study evaluating the effect of NaOH treatment on methane production from rice straw (Zhang et al., 2015). Hemicellulose having a molecular weight lower than that of cellulose is therefore easily degradable by the anaerobic microorganisms. Its solubilization is also elucidated by the disruption and the breakdown of hydrogen bonds by the alkaline solution (Hendriks and Zeeman, 2009). The lignin removal is due to the saponification of intramolecular ester bonds which crosslink hemicellulose and lignin (He et al., 2008). It was reported in 2010 that lignin degradation rate increased from 9% to 46% when corn stover was exposed to alkaline pretreatment at ambient temperature while the NaOH loading rate increased from 1.0% to 7.5% (Zhu et al., 2010). Regarding cellulose, it has shown in an alkaline medium the lowest removal rate between all three components; its native structure can be converted then to a more compact form, significantly limiting its reduction yield (Chen et al., 2014; He et al., 2008).

Alkaline treatment using NH_3 showed no effects on the degradability of the fibers (Fig. 4). In fact, NH_3 is a weak base because it does not completely dissociate into NH_4^+ and OH^- ions limiting its impact on the lignocellulosic matrix. Our results are consistent with those found in a previous study indicating no effects of NH_3 on methane yield and fibers biodegradability (Nordmann, 2013).

On another note, acid treatment using 10% HCl was only effective in removing hemicellulose displaying the highest reduction yield (79.4%) (Fig. 4). Generally, these pretreatments are known to improve degradation of xylans which are the main components of hemicellulose (Nizami et al., 2009). These findings are supported by microscope observations of hemicellulose structural modifications caused by acid pretreatment on sunflower oil cake (Monlau et al., 2013b). Despite the high hemicellulose reduction in GP, our study recorded a limited increase in methane production (+5%). This may be justified by the fact that the initial hemicellulose content in raw GP is low (around 7%,

Table 3
Van Soest fractionation of the raw GP and the treated GP after selected pretreatment conditions (freezing at −20 °C, 10% NaOH, 10% NH_3, 10% HCl, PEF Z_1, US Z_1 and 10% NaOH + freezing at −20 °C), based on VS.

	Fresh GP	F −20 °C	10% NaOH	10% NH_3	10% HCl	PEF Z_1	US Z_1	10% NaOH + F −20 °C
Soluble components ($g\,kg^{-1}$ of VS)	505 ± 15	506 ± 13	691 ± 17	513 ± 13	557 ± 15	507 ± 13	528 ± 14	688 ± 16
Hemicellulose ($g\,kg^{-1}$ of VS)	68 ± 2	67 ± 3	49 ± 1	67 ± 2	24 ± 1	68 ± 2	59 ± 1	48 ± 1
Cellulose ($g\,kg^{-1}$ of VS)	188 ± 4	189 ± 5	146 ± 4	185 ± 4	182 ± 3	187 ± 4	189 ± 4	148 ± 3
Lignin ($g\,kg^{-1}$ of VS)	239 ± 5	238 ± 4	114 ± 3	235 ± 5	237 ± 5	238 ± 5	224 ± 4	116 ± 2

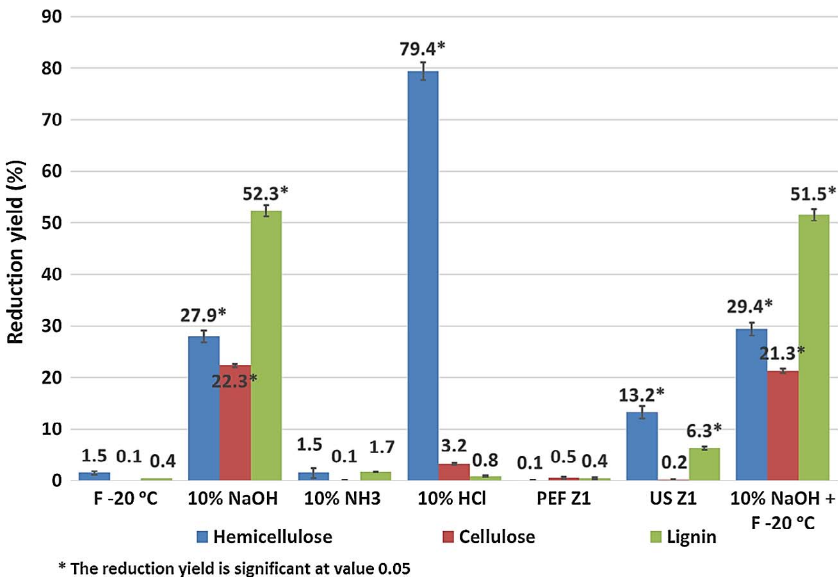

Fig. 4. Fibers reduction yield (%) after selected pretreatment conditions. Bars represent the average standard deviation of the experimental data.

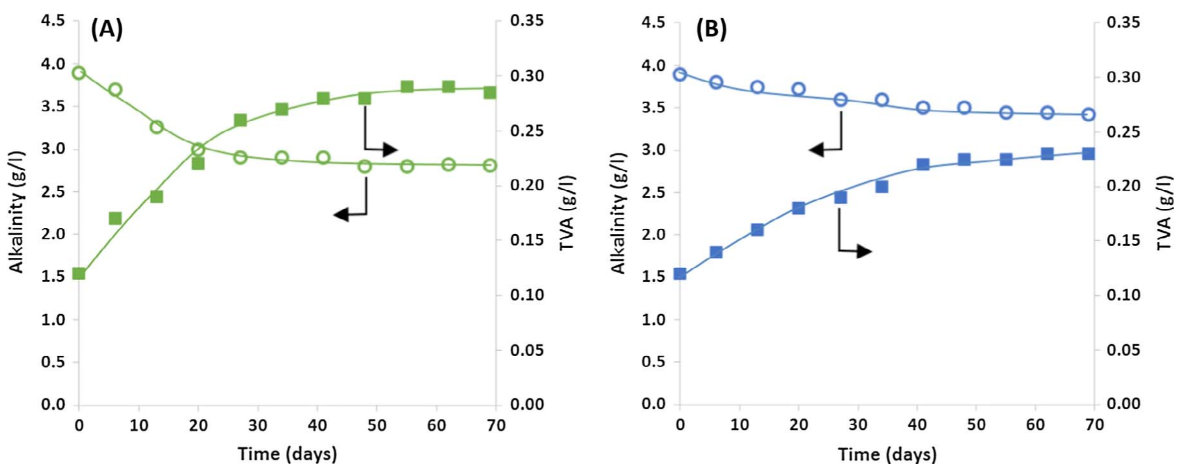

Fig. 5. Evolution of alkalinity and TVA/alkalinity ratio during the operation period. The values represent the means observed in the digesters containing untreated GP (A) and those containing GP treated with 10% NaOH (B).

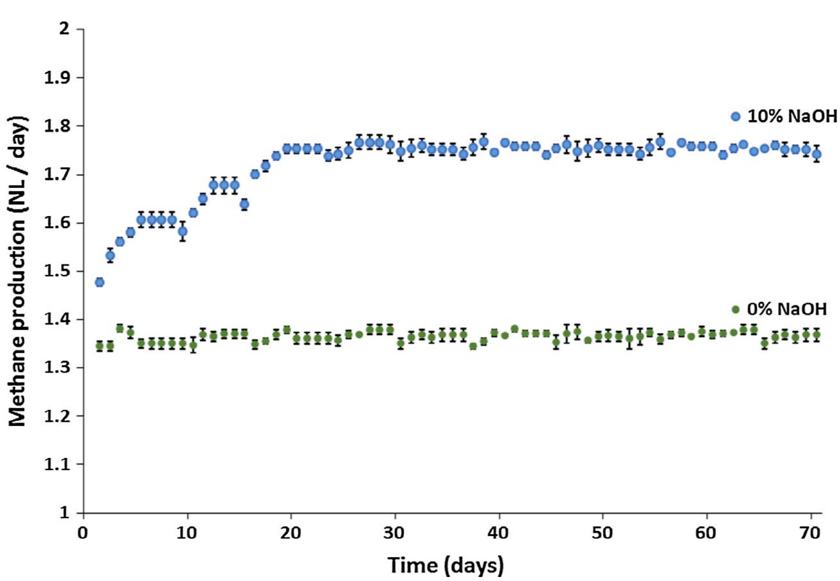

Fig. 6. Mean daily methane production during the operating period of the digesters containing untreated GP (0% NaOH) and those with GP treated with 10% NaOH. Bars represent the average standard deviation of the experimental data.

based on VS) (Table 3) and that lignin and cellulose accounting for almost 43% of the GP based on VS, are not affected by acidic conditions (Fig. 4).

Otherwise, PEF treatment did not affect the chemical composition of GP (Fig.4). Actually, it has been shown that the electroporation phenomena induced by PEF leads to a drastic increase in permeability, due to the appearance of pores in cell membranes releasing some components to the extracellular medium (Ben Ammar et al., 2011). For that reason, PEF has been widely used for the extraction of various compounds from GP without modifying the chemical composition of the treated samples (El Darra et al., 2016; Rajha et al., 2015). During AD process, pores created by PEF will facilitate the access to intracellular components which become more amenable to enzymatic hydrolysis. This can explain the obtained 4% improvement of GP methane production.

After sonication of GP, significant reduction yields of 13.2% and 6.3% were obtained, for hemicellulose and lignin, respectively (Fig. 4). In fact, US pretreatment at low frequency mechanically disrupts the cell structure and matrix leading to a reduction in particle size (Chu et al., 2002). Sonication creates small cavitation bubbles; hydrolyzing hemicellulose fractions into simpler reducing sugars readily biodegradable by microorganisms (Kumar and Sharma, 2017). As for lignin, our finding is consistent with another one showing that sonication of wheat straw increased delignification by a maximum of 8% compared to untreated samples (Sun and Tomkinson, 2002).

To end with the coupled treatment using 10% NaOH and freezing at −20 °C, the later did not improve fibers reduction yields compared to 10% NaOH alone (Fig.4). These results are expected since freezing does not specifically modify the fiber content; however, it leads to high cell deterioration, thus facilitating the degradation of the matter when in contact with hydrolytic microorganisms.

3.3. Effect of alkaline treatment on methane production in continuous digesters

Today, increasing gas production remains one of the main targets when converting biomass to methane. Altogether, our results highlight the positive impact of coupling alkaline treatment using 10% NaOH and freezing at −20 °C on GP AD. Accordingly, experiments conducted in batch mode must be validated in a scaled-up process using continuous digesters, to productive use by agricultural industries and wineries.

In continuous mode, the particle size of GP should not be too large, otherwise it would cause clogging of the digesters, mainly due to seeds presence. Additionally, we confirmed previously the positive effect of grinding on the anaerobic biodegradability of GP (El Achkar et al., 2016). For that, two sets of GP were prepared; the first represents the grinded GP without alkaline treatment and the second corresponds to the grinded GP treated with 10% NaOH. Following the alkaline pretreatment, the samples were ground to reach a particle size range between 0.04 µm and 1000 µm with a clear bimodal distribution centered on 25 µm and 500 µm, as evaluated using laser diffraction analysis (Mastersizer 2000, Malvern Instruments, United Kingdom). Homogeneous samples of each set were frozen at −20 °C. Prior to the continuous experiments, the effect of grinding was evaluated in batch mode and results showed that the cumulative methane yields were 0,208 $Nm^3\,kg^{-1}$ and 0,260 $Nm^3\,kg^{-1}$ of COD (data not shown), for the first and the second set of GP, respectively. Regarding the experiments in continuous mode, two digesters (1 and 2) were fed with the first set of GP and the two others (3 and 4) with the second.

For digesters (1) and (2) containing untreated GP, the mean pH of the medium decreased from 7.4 to 7.0 without any adjustment throughout the study. According to Fig. 5 (A), alkalinity decreased from 3.9 to 2.8 g eq. $CaCO_3\,L^{-1}$ and Total Volatile Acidity (TVA) increased from 0.12 to 0.29 g eq. $CH_3COOH\,L^{-1}$, by the end of experiments. We note that, an important criterion for confirming the digester stability is the TVA/alkalinity ratio. In this study, it increased from 0.03 to 0.1. This ratio affirms our digesters stability, with the reported normal ratio being less than 0.4 (Callaghan et al., 2002). Finally, the average methane production was stabilized at 1.37 ± 0.01 NL (Fig. 6).

For digesters (3) and (4) containing GP treated with 10% NaOH, a pH value of about 7.4 ± 0.05 was recorded throughout the experiments. Alkalinity decreased from 3.9 to a stabilized value of 3.5 g eq. $CaCO_3\,L^{-1}$ and TVA increased from 0.12 to 0.23 g eq. $CH_3COOH\,L^{-1}$, at the end of the experiments (Fig. 5, B). The corresponding TVA/alkalinity ratio increased from 0.03 to 0.07, indicating system stability. The mean methane production was stabilized at 1.75 ± 0.01 NL (Fig. 6); we conclude that alkaline treatment increases GP methane production by about 27%, in comparison to the untreated samples.

For a technical evaluation of the digester, the efficiency of GP methane production was determined by comparing the batch and continuous modes. Before and after the use of coupled pretreatment, methane production in the continuous digesters represents, in both cases, about 85% of that obtained with the batch system. In general, across anaerobic digesters operating with vegetable crops, it is possible to reach 70–90% of the methane potential in batch (Kafle and Kim, 2013; Zhang et al., 2013). Herein, the obtained yields stand for the success of our study on a large-scale test.

4. Conclusions

Among all tested pretreatments, alkaline treatment with NaOH is the most effective to improve GP hydrolysis rate and methane production by highly altering its lignocellulosic structure. Its combination with freezing treatment results in an increase in methane production of about 68% in batch mode, compared to the raw samples. Furthermore, the selected pretreatment conditions were validated on a larger scale, using continuous digesters. Although, some pretreatment conditions have been successful to some extent, the development of coupled pretreatments appears to be necessary to maximize the obtained effects, in a cost-effective and environmentally friendly manner.

Acknowledgement

The authors are grateful to the Research Council of Saint Joseph University of Beirut (Project FS66), The National Council for Scientific Research – Lebanon (CNRS-L), Arcenciel association and the French Institute in Lebanon for the financial support brought to the PhD work in the frame of which this study has been accomplished. We are also thankful to Yves Lemée and Paul Réthoré for providing the grape pomace and to Gael Le Noane for supplying the inoculum.

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